SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Deign **Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:20
Duration : 1 Hour

Scenario Based

Code of Conduct:

I A. Manoj Reddy 192311171 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Manoj

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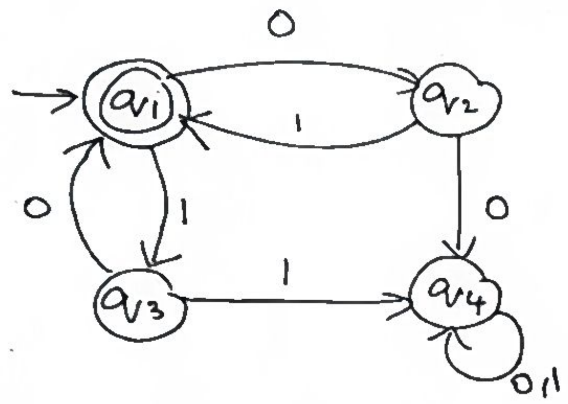
1.Given a DFA used for validating a specific PIN pattern bbbc, bca ,c, ca, caaaaaa minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

2.Demonstrate how a given finite automaton can be converted into a regular expression



3.Consider the following grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

C Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

4.I)Show that if L1 and L2 are regular then (L1+l2) also regular

Ii)Is the language { a^p| p is prime} regular? Justify

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandabl e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

88/100

DFA Minimization:

Given

$L = \{ bbbc, bca, cica, caaaaaa \}$

Alphabet $\Sigma = \{a, b, c\}$

State Transition table:

State / IP	a	b	c
A	x	B	F
B	x	C	E
C	x	D	x
D	x	x	J
E	J	x	x
F	H	x	x
G	H	x	x
H	I	x	x
I	E	x	J
J	x	x	x

The Minimized DFA states are:

Minimized States:

$\{ q\,abbc,\ q\,bca,\ q\,caaaaaa \},\ \{ q\,bc,\ q\,caaaaa \}$

$A = \{q_0\} \rightarrow$ initial, $B = \{q_1 b\}$ $C = \{q\,bb\}$

$E = \{q\,b,\ q\,aaaaa\}$, $I = \{q\,caac\}$, $J = \{q_1 b\}$

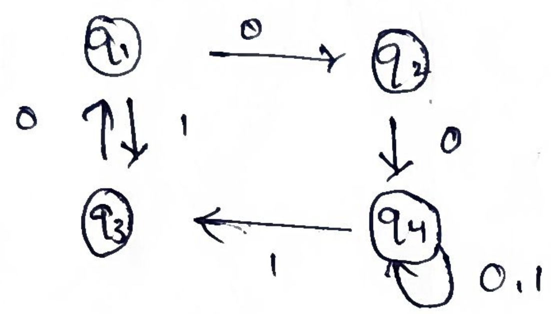
$E = \{q_0, q_{caaaaa}\}$, $I = \{q_{caac}\}$, $J = \{q_{accc}\}$

$F = \{q_c\}$, $G = \{q_{ca}\}$ $H = \{q_{caa}\}$

$x \rightarrow$ dead state

∴ Given PIN is valid

(2)



S_1:-

$q_1 = 0.q_2 + 1 + 1.q_3 + \epsilon$

$q_2 = 1.q_1 + 0.q_4$

$q_3 = 0.q_1 + 1q_4$

$q_4 = (0+1)q_4$

S_2:-

Eliminating Dead state (q_0)

Since q_4 has no path to the final state q_1 it

cannot produce accepting string.

$q_1 \rightarrow$ initial and final state

$q_4 \rightarrow$ is also a final state

$q_1 \rightarrow$ has a loop on a

from q_1, 1 can load q_4

$0^* + 0^* 1(a^* 1)(a+1)^*$

Grammar $S \to aSc \mid T \mid U \mid c$

$T \to bTc \mid \epsilon$

$U \to aUb \mid \epsilon$

(a) language

$T \Rightarrow b^m c^m$

$U \to a^m b^m$

$\therefore$ generates $a^n b^i c^{i+j}$ $\cup a^j b^j c^i$

where $i, j \geq 0$

(B) derivation tree for aabbc

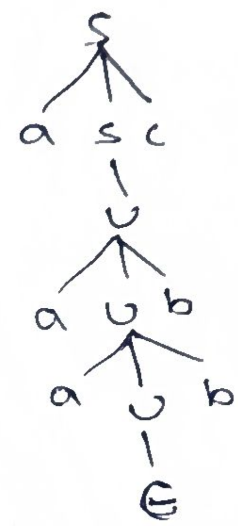
$S \to aSc$, $S \to U$

$S \to aUc$, $U \to aUb$

$S \to aaUbc$, $U \to aUb$

$S \to aabbcU$, $U \to \epsilon$

$S \to aabbc\epsilon$



~~Ambiguity~~

$S \to \epsilon$

$S \to T \to \epsilon$

$S \to U \to \epsilon$

$\therefore$ Grammar is ambigous

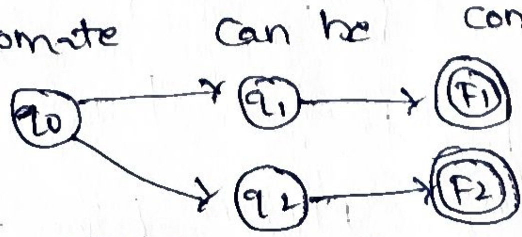
(4)

Regular languages:-

(i) prove $L_1 + L_2$ is Regular

let L_1 and L_2 be regular

Lies automate can be combined using initial



the resulting NFA accepts $L_1 \cup L_2$, is regular

Assume it is regular with pumping length (base prime $P > M$) $w = a^P$

let $y = a^k$, $k > 0$

Pump y,

$p+1$ times : $\{x y^{p+1} z\} = p(1+k)$

This is composite,

Not prime so key contracts

$\{a^p \mid p$ is prime$\}$ is not regular$_{\shortparallel}$